20/3/2026 DAY-5

BLOCK-1 : Aptitude

1) A train travels 60 km in 1 hour, what is its speed in m/s?

Solu:

60 km/h

km/h → m/s

$km/h \times \frac{5}{18}$

$\frac{60 \times 5}{18} = \frac{50}{3}$ = 16.66

$3 \overline{)50}$ = 16.6

= 16.66 m/s

2) A car travels at 40 km/h for 2 hours and 60 km/hr for next 2 hours. Find average speed.

Solu:

Total Distance = (40×2) + (60×2)

= 80 + 120 = 200 km

Total time = 2 hours + 2 hours

= 4 hours.

Average speed = $\frac{200}{4}$ = 50 km/hr

3.) Two trains moving in opposite directions at 50 Km/hr and 70 Km/hr. Find relative speed.

Solu:- Opposite directions = 50 + 70

= 120 Km/hr.

Relative speed = 120 Km/hr.

4.) A person walks at 5 Km/hr and runs at 10 Km/hr. Find the time taken to cover 15 Km if half distance is walked and half is run.

Solu:

Total distance = 15 Km.

Half = 15/2 = 7.5 Km.

Walking => $\frac{7.5}{5}$ = 1.5 hr.

Running => $\frac{7.5}{10}$ = 0.75 hr.

Total time = 1.50 + 0.75 hr.

= 2.25 hrs.

$$\begin{array}{r} 1.50 \\ 0.75 \\ \hline 2.25 \end{array}$$

5.) If a train crosses a pole in 10 seconds and its length is 100 meters, find its speed.

Solu: Speed = $\frac{\text{Distance}}{\text{Time}}$ => $\frac{100\,m}{10\,s}$ = 10 m/s

m/s => Km/hr => $\frac{18}{5}$

$10 \times \frac{18}{5}$ = 36 Km/hr.

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Block -2 : Verbal Ability.

1.) why are algorithms important?

Algorithms help solve problems efficiently and improve system performance.

2.) What happens if inefficient algorithms are used?

They increase execution time and use more resources.

3.) What is optimization in computing?

Optimization means improving performance by reducing time and resource usage.

Grammar Correction:

1.) He have been working here since 2 years.

He has been working here for 2 years.

2.) She did not wrote the code.

She did not write the code.

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3) They is preparing for interviews.
They are preparing for interviews.

Block-3 Communication Practice:

Sliding window

Sliding window is a technique used in programming to reduce time complexity. Instead of checking all possible cases, we use a window and move it step by step.

For Example: in the problem "Longest substring without repeating characters", we check char one by one. if a duplicate comes, we remove characters from the left side and continue. This make the solution faster.

In real-world applications, sliding window is used in data streaming, network monitoring, and substring problems. It helps to process large data efficiently.

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